

The Pais–Uhlenbeck Oscillator: Canonical Signs, Resonant Tori, Jordan Collision, and a Quantum Difference Spectrum

HCS-C359 source-local theorem package

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Abstract

For the two-frequency Pais–Uhlenbeck oscillator we prove an exact Ostrogradsky normal form and classify every classical orbit closure. A rational frequency ratio closes every orbit; an irrational ratio leaves only the equilibrium and single-mode circles periodic, while each genuine double-mode orbit is dense on a two-torus. We then resolve the equal-frequency Jordan collision and all real sign degenerations. Finite exact receipts audit conventions; the continuum statements are proved analytically.

1 Frozen model and canonical sign

Consider

$$L = \frac{1}{2}\{\ddot{x}^2 - (\omega_1^2 + \omega_2^2)\dot{x}^2 + \omega_1^2\omega_2^2x^2\}, \quad 0 < \omega_1 < \omega_2. \quad (1)$$

The higher-order Euler–Lagrange equation factors as

$$(D^2 + \omega_1^2)(D^2 + \omega_2^2)x = 0. \quad (2)$$

With $q_0 = x$, $q_1 = \dot{x}$, $p_1 = \ddot{x}$ and $p_0 = -(\omega_1^2 + \omega_2^2)\dot{x} - x^{(3)}$, the Ostrogradsky Hamiltonian is

$$H = p_0q_1 + \frac{1}{2}p_1^2 + \frac{1}{2}(\omega_1^2 + \omega_2^2)q_1^2 - \frac{1}{2}\omega_1^2\omega_2^2q_0^2. \quad (3)$$

Put $\Delta = \omega_2^2 - \omega_1^2$ and define

$$\begin{aligned} Q_1 &= (p_1 + \omega_2^2q_0)/\sqrt{\Delta}, & P_1 &= (p_0 + \omega_1^2q_1)/\sqrt{\Delta}, \\ Q_2 &= (p_1 + \omega_1^2q_0)/\sqrt{\Delta}, & P_2 &= -(p_0 + \omega_2^2q_1)/\sqrt{\Delta}. \end{aligned} \quad (4)$$

Direct Poisson-bracket calculation gives $\{Q_j, P_k\} = \delta_{jk}$ and zero cross brackets. Substitution gives

$$H = -\frac{1}{2}(P_1^2 + \omega_1^2Q_1^2) + \frac{1}{2}(P_2^2 + \omega_2^2Q_2^2). \quad (5)$$

Thus our convention has a negative low-frequency mode and a positive high-frequency mode. This explicit check is the *canonical-sign owner*; changing either sign changes the theorem.

Theorem 1 (Complete classical closure). *In the chamber above, the linear Hamiltonian flow is complete. If $\omega_1/\omega_2 \in \mathbb{Q}$, every trajectory is periodic. If the ratio is irrational, exactly the equilibrium and single-mode trajectories are periodic, and each double-mode trajectory is dense on its invariant two-torus.*

Proof. Equation (2) has the general solution

$$x = a_1 \cos \omega_1 t + b_1 \sin \omega_1 t + a_2 \cos \omega_2 t + b_2 \sin \omega_2 t. \quad (6)$$

Equivalently, (5) preserves the two modal radii $R_j^2 = Q_j^2 + P_j^2/\omega_j^2$. If $\omega_1 = gm, \omega_2 = gn$ with coprime positive m, n , then $T = 2\pi/g$ advances the two phases by $2\pi m, 2\pi n$. It is a common period; a collapsed one-mode circle can have a smaller least period.

For irrational ratio, a periodic double-mode point would require both $\omega_j T \in 2\pi\mathbb{Z}$, contradicting irrationality. Single-mode circles retain period $2\pi/\omega_j$. Kronecker’s theorem makes the two-frequency phase orbit dense in $S^1 \times S^1$, and the invertible map (4) carries that closure to the Ostrogradsky torus. Constant coefficients give completeness for all real time. QED

2 Collision and sign boundaries

The normal map (4) is singular at collision, so the boundary must be solved from (2), not inferred by continuity.

Proposition 1 (Complete real factor atlas). *For the real-factor equation $(D^2 + \alpha)(D^2 + \beta)x = 0$, the distinct positive chamber is supplemented by the following exhaustive faces, up to interchange of α, β .*

1. If $\alpha = \beta = \omega^2 > 0$, then

$$x = (a + bt) \cos \omega t + (c + dt) \sin \omega t. \quad (7)$$

The first-order matrix has one size-two Jordan block at each of $\pm i\omega$; periodicity holds exactly when $b = d = 0$.

2. If $\alpha = 0 < \beta = \omega^2$, the basis is $1, t, \cos \omega t, \sin \omega t$; boundedness and periodicity hold exactly when the coefficient of t vanishes. If $\alpha = \beta = 0$, the basis is $1, t, t^2, t^3$, and only its constant subspace is bounded or periodic.
3. A negative factor $-\nu^2$ contributes $e^{\nu t}, e^{-\nu t}$. Pairing it with a positive factor, a zero factor, or a distinct negative factor supplies the corresponding direct union of bases. At the repeated negative face, the basis is $e^{\nu t}, te^{\nu t}, e^{-\nu t}, te^{-\nu t}$. Every nonzero hyperbolic component is unbounded in at least one time direction. Hence the bounded/periodic subspace is the oscillatory plane for positive/negative factors, the constant line for zero/negative factors, and zero for two negative factors, repeated or distinct.

Proof. Each item is the kernel basis of its factored constant-coefficient polynomial. At the positive collision the characteristic polynomial is $(\lambda^2 + \omega^2)^2$, while the nullity of the first-order matrix at each root $\pm i\omega$ is one. Algebraic multiplicity two and geometric multiplicity one force the asserted length-two Jordan blocks. The listed positive, zero, and negative cases exhaust both real factors. QED

This direct calculation is the *Jordan-boundary owner*. It prevents the imaginary repeated roots from being misreported as a diagonalizable stable oscillator.

Source and claim boundary

Pais and Uhlenbeck introduced the non-local-action model in *Physical Review* 79 (1950), 145–165, doi:10.1103/PhysRev.79.145. Smilga analyzed higher-derivative ghost dynamics in *Nuclear Physics B* 706 (2005), 598–614, doi:10.1016/j.nuclphysb.2004.10.037. Bolonek and Kosiński studied Hamiltonian structures in arXiv:quant-ph/0501024. These sources establish lineage; every displayed claim above is rederived for the frozen convention. No priority claim is made.